

ALHASSAN ABDULAI

Manchester, United Kingdom | +44 7826 448718 | abdulaia549@gmail.com | linkedin.com/in/alhassan-abdulai-1a9198169 | github.com/Authenticscholar/peak-cluster-ccs-analysis | Available to relocate to Amsterdam · EU mobility

RESEARCH PROFILE

Environmental and climate researcher focused on physical climate risk, drought and flood exposure, and adaptation options that reduce societal risk. Dual MSc training (Erasmus Mundus Environmental Science, Policy and Management — Distinction, GPA 3.73; Climate Change Engineering) with hands-on climate-projection analysis (CMIP5), GIS-based flood and drought risk mapping, and reproducible Python modelling. Peer-reviewed publications on climate change hotspots (rainfall, temperature and drought trends) and on climate adaptation and disaster risk reduction in Africa. Motivated to develop multi-risk simulation methods for consecutive droughts and compound flood–drought risk in Europe, and to assess how adaptation behaviour reduces risk under alternative climate scenarios — aligned with IVM Water & Climate Risk and the GEB modelling framework.

KEY QUALIFICATIONS FOR THIS PHD

- Climate risk evidence: co-author of peer-reviewed drought-trend hotspot analysis (GIS & statistical modelling) and a comprehensive review of climate adaptation and disaster risk reduction in Africa; Stanford Global Sustainability Challenge Top 6 (Adaptation & Resilience).
- Climate-model and geospatial data: processed CMIP5 RCP 4.5 projections (1950–2100) for temperature, precipitation and dryness across Africa (University of the Aegean poster); ArcGIS Model Builder and Python pipelines; QGIS flood-projection styling and flood susceptibility layer preparation (TWI, drainage density, slope, precipitation, LULC, soil).
- Scientific Python: built a reproducible research pipeline (numpy, pandas, matplotlib) for Monte Carlo simulation (N=10,000), scenario and sensitivity analysis (github.com/Authenticscholar/peak-cluster-ccs-analysis); strongly motivated to deepen Python skills and learn HPC workflows for large-scale GEB simulations.
- Adaptation and societal vulnerability: drought policy analysis for northern Ghana food security; desertification mapping (Digital Earth Africa); community-based flood risk management literature synthesis (Algeria); field RCT operations with IPA linking household vulnerability to programme response.
- Interdisciplinary collaboration: delivery across UNFCCC, SEforALL, Common Fund for Commodities and multi-university Erasmus Mundus consortia; fluent academic English; experience presenting and publishing scientific results.

RESEARCH INTERESTS

- Multi-hazard and consecutive drought risk assessment at regional and local scales
- Compound flood–drought interactions and sectoral vulnerability under climate scenarios
- Adaptation measures and behavioural responses that reduce climate risk (policy-to-society link)
- Large-scale climate and geospatial data processing for comparable climate risk assessments
- Integrating hydrological / flood hazard information with societal impact and adaptation modelling

EDUCATION

MSc Environmental Science, Policy and Management — Erasmus Mundus

Central European University · University of the Aegean · University of Manchester | 2024 – 2026

Distinction (GPA 3.73). Coursework: Quantitative Methods, GIS / geospatial analysis, Energy Markets, Energy Transition, Environmental Policy, EIA, Pollution Management. Thesis: Path dependency and cement-industry decarbonisation (Peak Cluster CCS) — Monte Carlo LCO₂A, MAC/ETS scenarios, policy lock-in analysis (Supervisor: Prof. Ian Kane).

MSc Climate Change Engineering

Université Abou Bekr Belkaid — Pan African University Institute of Water and Energy Sciences including Climate Change (PAUWES) | 2022 – 2024

Coursework: Climate Change Mitigation, Climate Modelling, Low Carbon Climate Resilient Development, Transdisciplinarity in Climate Change. Thesis: Blockchain and carbon-market transparency in Africa. Co-authored abstract on community-based flood risk management in Algeria.

BSc Agribusiness (Climate Change & Food Security)

University for Development Studies, Ghana | 2015 – 2019

First Class Honours · CGPA 4.51 · Best Graduating Student (department). Econometrics, statistics, sustainable agriculture, climate change and food security.

SELECTED RESEARCH RELEVANT TO CLIMATE RISK & ADAPTATION

Climate change hotspots — rainfall, temperature and drought trends (Africa)

Co-author, Journal of Earth System Science (2026). GIS and statistical modelling of climate hotspots with explicit drought-trend analysis — directly relevant to consecutive drought risk characterisation.

Climate adaptation and disaster risk reduction (Africa)

Co-author, Journal of Water and Climate Change (2025). Comprehensive review of adaptation and DRR evidence — framing how measures reduce risk under climate stress.

CMIP5 climate-projection analysis — Africa under RCP 4.5 (1950–2100)

Research poster, University of the Aegean (with Ilyas & Kontos). Mapped spatial and temporal shifts in temperature, precipitation and dryness using Köppen-Geiger classification and trend detection (temperature $R^2=0.93$; dryness $R^2=0.66$); automated ArcGIS Model Builder and Python folder/pipeline structure for RCP 4.5 / 8.5 outputs.

Flood susceptibility GIS layers and flood risk synthesis

Prepared multi-criteria flood classification layers (topographic wetness index, elevation, slope, precipitation, LULC, NDVI, drainage density, soil type) and contributed to a community-based flood risk management review for Algeria (PAUWES).

Drought, food security and adaptation options — northern Ghana

Policy analysis on consecutive drought threats to agriculture-dependent livelihoods; Digital Earth Africa GIS mapping of desertification; sector climate-impact synthesis using CMIP5 / USAID climate risk evidence for Ghana.

Reproducible quantitative modelling (Python + Stata)

MSc thesis pipeline: Monte Carlo simulation, tornado/sensitivity analysis, ETS scenarios; optional R cross-check of Monte Carlo median. Demonstrates scientific coding discipline transferable to GEB model development and experiment design.

RESEARCH & PROFESSIONAL EXPERIENCE

Climate & Carbon Markets Consultant

Common Fund for Commodities (Amsterdam) & Forestry and Land Check | Remote | Jul 2025 – Oct 2025

- Produced GIS-based environmental analysis and risk assessments for five project locations.
- Designed MRV / monitoring plans for REDD+ and PES covering 10,000+ hectares; synthesised 20+ technical reports into KPIs for a USD 75M outcomes fund.
- Collaborated with international partners on climate finance and land-use risk documentation.

High Integrity Project Development Officer

Sustainable Energy for All — Africa Carbon Markets Initiative | Ghana / Remote | Mar 2024 – Oct 2024

- Supported GHG baselining, MRV design and methodology research (including diesel genset displacement for DRE); engaged 50+ project developers across Africa.
- Contributed to Mission 300 programme activities and Carbon Hub intelligence synthesis.

Intern — Climate Transparency and Carbon Markets

UNFCCC Regional Collaboration Centre for East and Southern Africa | Remote | Jul 2023 – Feb 2024

- Analysed 15+ national GHG inventories and Biennial Update Reports; identified data gaps and methodological inconsistencies under the Enhanced Transparency Framework.

- Produced structured synthesis on Article 6 and carbon pricing for regional processes.

National Climate Action Fellow

Youth Challenge International — HerStart | Ghana | Feb 2021 – May 2022

- Delivered climate action training to 300+ youth; facilitated 15 multi-stakeholder workshops (teaching and supervision experience relevant to the 10% teaching component).
- Reviewed 20 national climate policy documents and supported adaptation-oriented entrepreneurship mentoring.

Project Assistant

Innovations for Poverty Action — Escaping Poverty Project | Ghana | Sep 2019 – Feb 2021

- Supported multi-site randomised controlled trial operations; supervised field auditors and agents; managed tracking systems for large participant cohorts — experience with household vulnerability data relevant to agent-level societal response modelling.
- Applied Stata / Excel discipline for monitoring and operational data integrity.

PUBLICATIONS

- Saber, M., Posite, V. R., Abdulai, A., et al. (2026). Climate change hotspots in Africa: rainfall, temperature and drought trends (GIS & statistical modelling). *Journal of Earth System Science*, 135, 187.
<https://doi.org/10.1007/s12040-026-02867-4>
- Saber, M., Koroma, A., Posite, R. V., Ly, A., Tejan Bah, A. M., Abdulai, A., et al. (2025). A comprehensive review of climate change adaptation and disaster risk reduction in Africa. *Journal of Water and Climate Change*, 16(5), 1831–1862.
- Siddique, H. A., Kanagie, A., & Abdulai, A. (2024). Echoes of the past: Power and colonial impact on contemporary climate politics. In *Climate change and environmental politics* (pp. 141–166). Transnational Press London.

TECHNICAL SKILLS

Programming & quantitative methods: Python (numpy, pandas, matplotlib — reproducible research pipelines); Stata; SPSS; advanced Excel; Monte Carlo, scenario and sensitivity analysis; motivated to advance Python for agent-based / HPC modelling

Climate & geospatial data: CMIP5 climate projections; ArcGIS (incl. Model Builder); QGIS; Digital Earth Africa; flood susceptibility multi-criteria layers; drought / aridity trend analysis

Domain: Climate risk and adaptation; drought and flood risk evidence; disaster risk reduction; sectoral vulnerability (agriculture / food security); MRV and climate policy synthesis

Collaboration & communication: Scientific writing and peer-reviewed publication; conference/poster presentation; workshop facilitation; international project partner coordination

AWARDS & HONOURS

- Stanford Global Sustainability Challenge — Top 6 · Jury Award for Strategic Thinking, Adaptation & Resilience (2026)
- Schneider Electric Go Green Challenge — Global Winner · technical lead (2025)
- Erasmus Mundus Scholarship (2024)
- Geneva Challenge — 2nd Place globally, 1st in Africa (2023)
- PAU African Union Scholarship (2022)

REFERENCES

Reference letters can be requested; contact details for two academic/professional referees are listed below (VU application allows upload or contact details).

- Prof. Ian Kane — Thesis supervisor, University of Manchester (Peak Cluster CCS MSc thesis). Contact on request / via University of Manchester.

- Ishmail Azindoo Baako — Regional Technical Manager, West Africa, Innovations for Poverty Action · iabaako@poverty-action.org · +233 24 916 8102
- Dr Abdul Rahman Awal — Senior Lecturer, University for Development Studies · lawalrahman108@gmail.com · +233 20 402 4398